Answers and talk notes

Write and Represent Inequalities

What your student is learning

Tonight we show an inequality on a number line — and the open or closed circle tells whether the boundary number counts.

Answers

Ask your student to explain before you look.

1. **B.** 4 is a solution. 4 is a solution, because a closed circle means the number is included.
2. **C.** $x > 6$. $x > 6$ needs an open circle, because 6 is not a solution.
3. **A.** $x \leq 9$. $x \leq 9$ needs a closed circle, because 9 is included.
4. **D.** Right, toward larger numbers. Right, toward larger numbers, because x is greater than 2.
5. **B.** $x \leq 5$. $x \leq 5$, because 5 is included and the shading goes toward smaller numbers.

Talk about it

Does the number at the circle count? How does the circle show that?

Home mission

Use a ruler as a number line. Say: "No more than 12 inches." Put a coin on 12. Should the coin count as a closed or open circle? Which way do you slide your finger?

When they are stuck, ask

- Does the boundary number itself count as a solution?
- Which words tell you that — "at most", "more than"?
- Which direction do the allowed values go?

Try not to say

- "At least" means greater than, so use >.
- Use an open circle, it looks cleaner.
- An inequality has one answer like an equation.